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Quality Assurance in Adaptive Radiotherapy

Bin Cai, PhD, DABR

Mayo Clinic Florida



Disclosure

- None



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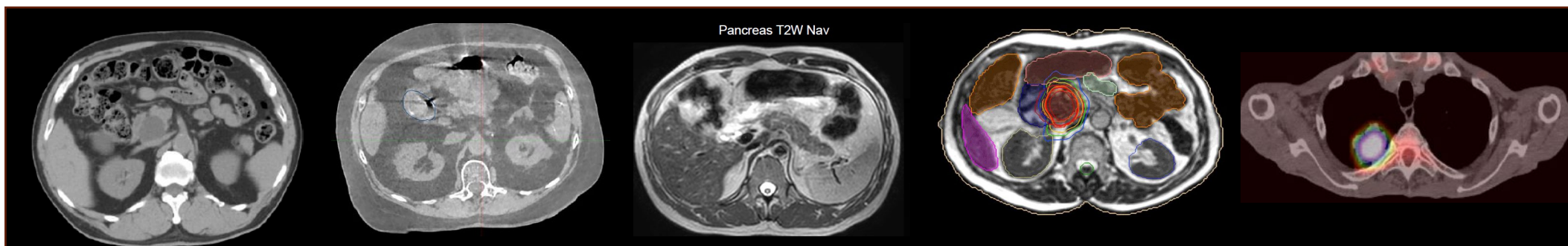
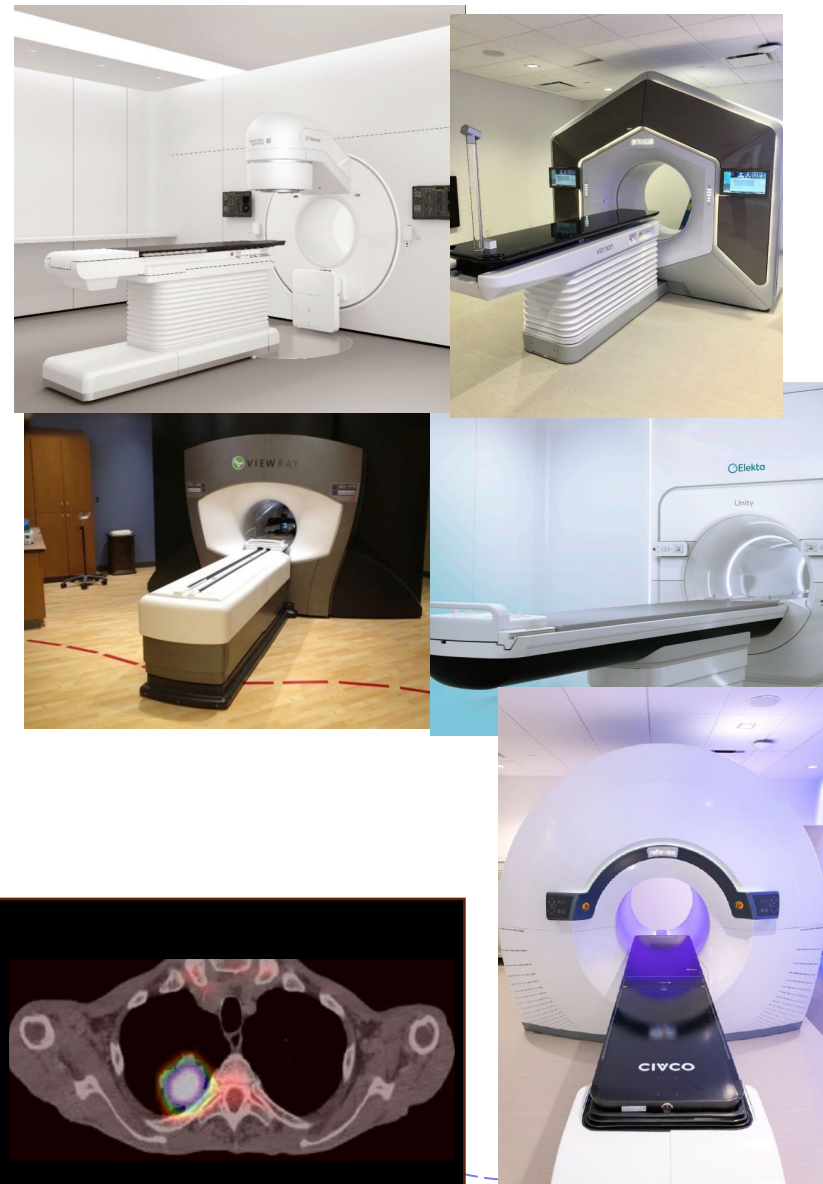
Learning Objectives

- Explain the unique QA characteristics in ART
- Describe the framework of ART QA
- Differentiate: System-level and Patient-specific QA
- Demonstrate QA strategies with examples

Why online ART became feasible now?

■ Enabling Technologies:

- High-quality onboard imaging -> visualization or detection of daily anatomical or functional changes
- Dedicated adaptive TPS -> recontouring, re-optimization and plan generation within minutes
- Rapid QA systems -> validate new plans prior to or immediate after delivery



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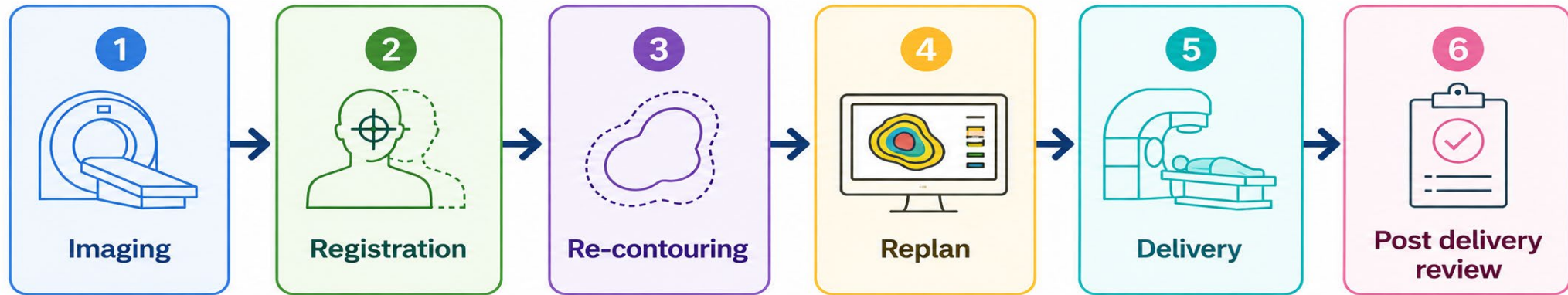
Why QA matters in ART?

- ART introduces daily or on-table plan modifications
- Workflow complexity is significantly higher than conventional RT
- QA must evolve from static verification → dynamic workflow assurance
- Errors may occur during each key ART step

ART QA is both technical and process-driven.

ART workflow and QA integration

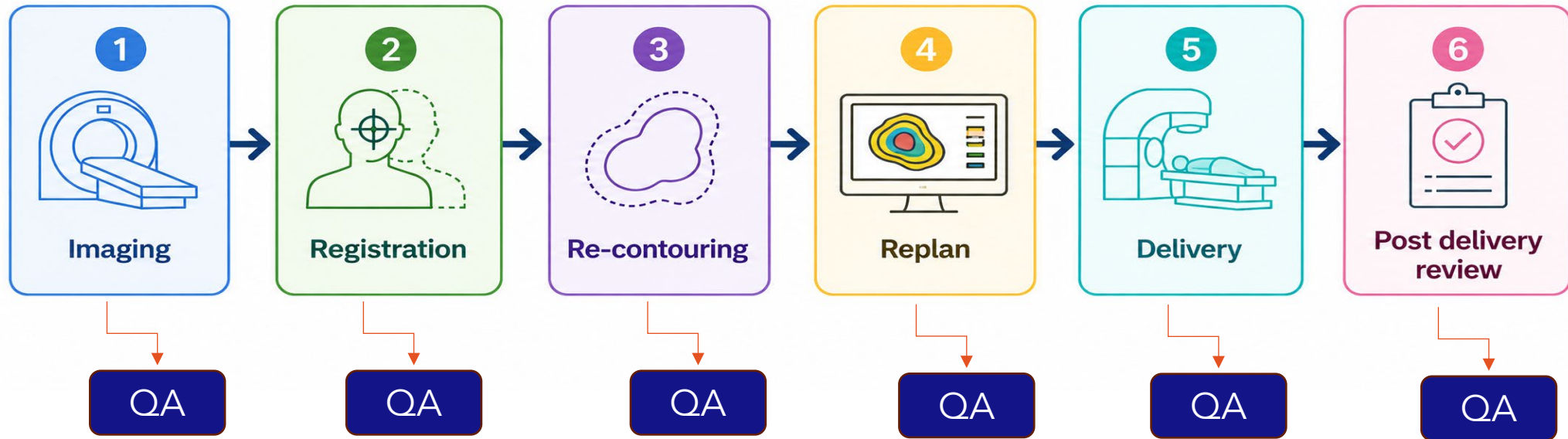
- A typical ART workflow



Discussion: What could go wrong in each ART step? Which step need QA?

ART workflow and QA integration

- A typical ART workflow



QA is embedded throughout the workflow

Adaptive vs Conventional RT QA

Conventional RT	Adaptive RT
Static plan	Dynamic adaptation
Offline QA	Online/time-sensitive QA
Plan-centric	Plan and Workflow-centric
Limited anatomy change	Frequent anatomy and/or functional variations

QA paradigm (and mindset) shift !

Unique Characteristics of ART QA?

ART QA requires

- Real-time validation
- Workflow integration
- Rapid decision making
- Interdisciplinary coordination
- Continuous verification

ART QA requires:



Real-time
validation



Workflow
integration



Rapid
decision making



Interdisciplinary
coordination



Continuous
verification

What are the goals of ART QA?

Four Core Goals



Accuracy



Safety



Efficiency



Consistency



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ART QA framework

The QA framework for ART can be divided into two interrelated categories:

System-level QA

- Commissioning and validation of adaptive platforms (e.g., machine, imaging systems, and TPS).
- Workflow and E2E testing—simulation of the entire adaptive process, ideally using phantoms with anatomical changes that trigger adaptation, to confirm functional readiness.
- Ongoing QA to maintain the reliability of imaging, TPS, and the treatment unit.

Patient-specific QA

- QA checks embedded in the daily ART process.
- Robustness of the initial plan and suitability for adaptation.
- Imaging QA (e.g., CT, CBCT, MRI, PET) and registration fidelity (rigid and deformable).
- Contouring and segmentation QA (manual, automated, or artificial intelligence (AI)-based, with uncertainty assessment).
- Plan adaptation, reoptimization, and plan review under time constraints.
- Treatment delivery QA (pretreatment verification, real-time monitoring, and posttreatment log-file analysis).

Unique QA demands

- E2E testing is essential, extending beyond beam output and TPS accuracy to the entire adaptive workflow.
- Dynamic phantoms or test datasets with simulated anatomical changes should be used to validate adaptation triggers and confirm workflow integrity.
- Process QA is a distinct requirement: QA and QC approaches should follow a FMEA framework: identifying the most vulnerable workflow steps, QA can be targeted to mitigate the highest-risk failure modes.

Overall Machine and TPS Commissioning

- Follow standard guideline and find available resources for specific system (if available).

Table 14.2 American Association of Physicists in Medicine Guidance documents for commissioning and quality assurance of adaptive radiotherapy equipment, indicating relevant documents pertaining to each modality(Chetty, Cai et al. 2024).

Reference	Description	Offline	X-ray	MRI	BgRT	Proton
TG-106(Das, Cheng et al. 2008)	Accelerator beam data commissioning equipment and procedures	X	X	X	X	
MPPG 5b(Geurts, Jacqmin et al. 2022)	Commissioning and QA of treatment planning dose calculations (MV photon and electron beams)	X	X	X	X	
TG-119(Ezzell, Burmeister et al. 2009)	IMRT commissioning: Planning and dosimetry comparisons	X	X	X	X	
TG-120(Low, Moran et al. 2011)	Dosimetry tools and techniques for IMRT	X	X	X	X	
TG-248(Walz-Flannigan, Weiser et al. 2019)	Interoperability Assessment for the Commissioning of Medical Imaging Acquisition Systems	X	X	X	X	X
MPPG 2b(McCullough 2021)	Commissioning and QA of X-ray-based IGRT systems	X	X		X	X
TG-104(Yin, Wong et al. 2010)	Role of In-Room kV X-Ray Imaging for Patient Setup and Target Localization	X	X		X	X
TG-142(Klein, Hanley et al. 2009)	QA of medical accelerators	X	X	X	X	
MPPG 8a(Smith, Balter et al. 2017)	Linear accelerator performance tests	X	X	X	X	
TG-53(Fraass, Doppke et al. 1998)	QA for clinical radiotherapy treatment planning	X	X	X	X	X
TG-105(Chetty, Curran et al. 2007)	Clinical implementation of Monte Carlo-based photon and electron external beam treatment planning	X		X		
TG-179(Bissonnette, Balter et al. 2012)	QA for IGRT utilizing CT-based technologies	X	X		X	X
MPPG 9a(Halvorsen, Cirino et al. 2017)	SRS-SBRT practice guideline		X	X	X	

More guideline for ART platform

- NRG ART consideration (2021), ASTRO safety white paper on ART (2024).
- AAPM Task group

Task Group No. 384 - Clinical implementation of automated segmentation for adaptive radiation therapy (ART) (TG384)

Task Group No. 395 - CT Guided Online Adaptive RT: Guidelines for quality assurance and clinical implementation (TG395)

Task Group No. 441 - CT based Adaptive Proton Therapy (APT): Guidelines for clinical implementation and quality management (TG441)

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Activities

1. To provide an overview of clinical experiences and challenges associated with various offline and online proton adaptive techniques.
2. To assess and establish the minimal requirements for volumetric imaging in adaptive proton therapy, including a review of in-room imaging modalities, geometric and imaging quality requirements, proton stopping power calculation accuracy, adaptation workflow, and quality management,
3. To investigate and recommend the staffing level and composition

Task Group No. 442 - Clinical implementation of Online Adaptive Radiotherapy (ART) on MR-guided Radiotherapy (MRgRT) systems (TG442)

Task Group No. 443 - Safe practice of adaptive treatment management and dose reporting (TG443)

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Activities

1. Survey of ART Treatment Strategies and Dose Reporting Tools: Conduct a comprehensive survey of ART treatment strategies and dose reporting tools currently offered by vendors and utilized clinically by users. This survey will provide valuable insights into existing practices and identify areas that require improvement.
2. Development of Safe Practice Guidelines for ART Treatment Management: Describe and recommend safe practices for ART considerations for MR-guided online adaptive RT.
6. Provide guidance on patient positioning, imaging, localization, segmentation, electron density propagation, treatment planning, plan quality review, and quality assurance considerations for MR-guided online adaptive workflow.

By-Laws: Not Referenced.

Rules: [Not Referenced.](#)

Standing? This is not a standing committee.

Approved Date(s): 1/1/2025 - 12/31/2028

Reviewable

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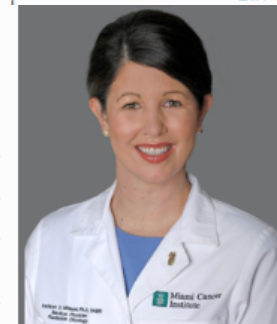
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Chairs

Chair



Bin Cai



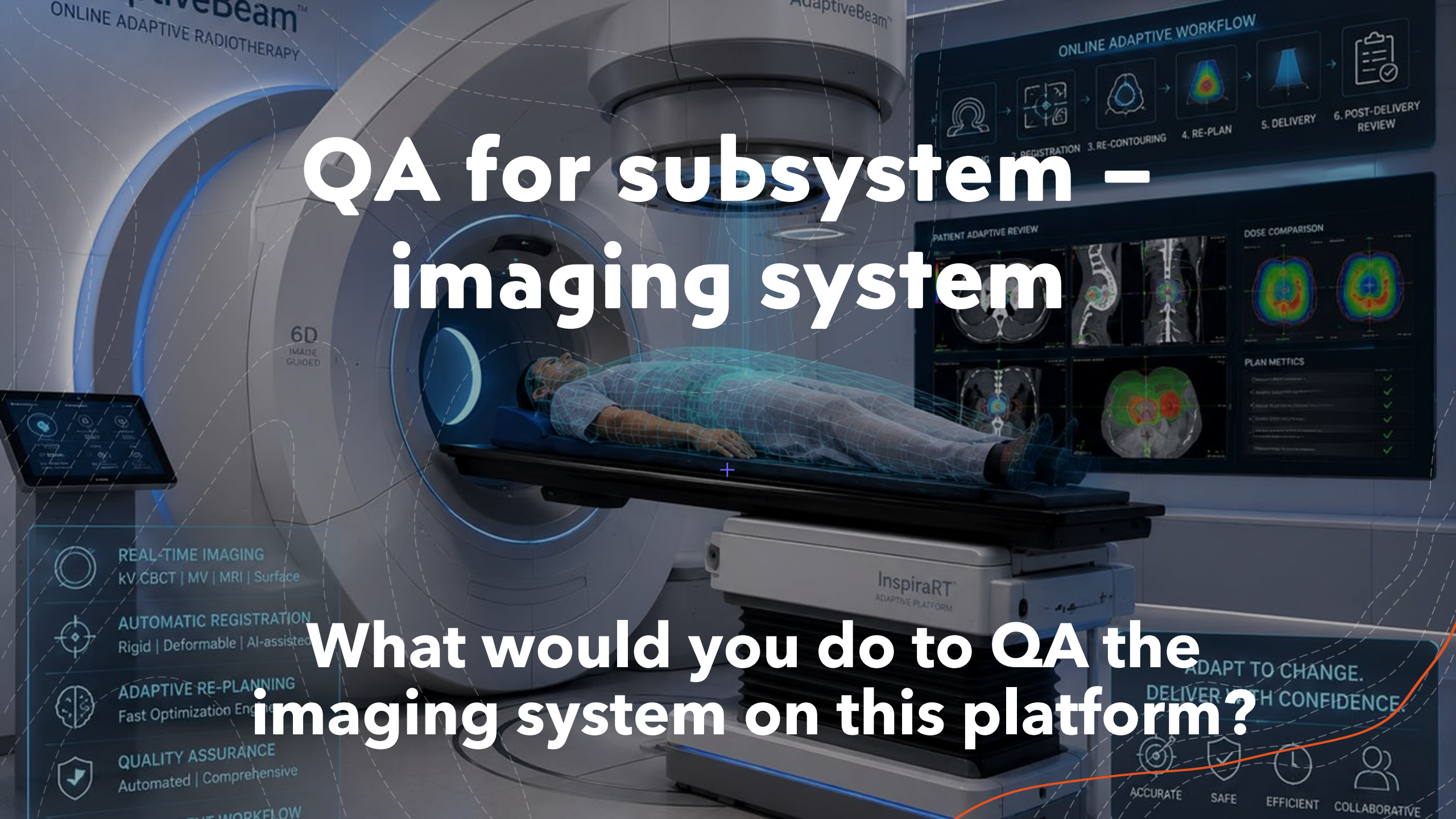
Kathryn Mittauer

QA for subsystems of ART platform

- Treatment planning system
- Treatment delivery system
- Online recontouring/replanning system
- Treatment monitoring and motion management
- Secondary dose calculation
- Others

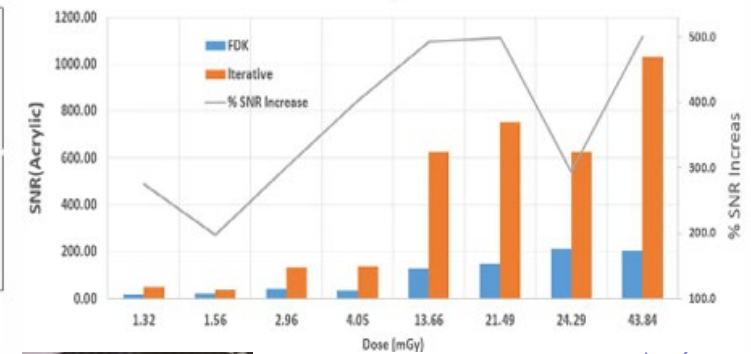
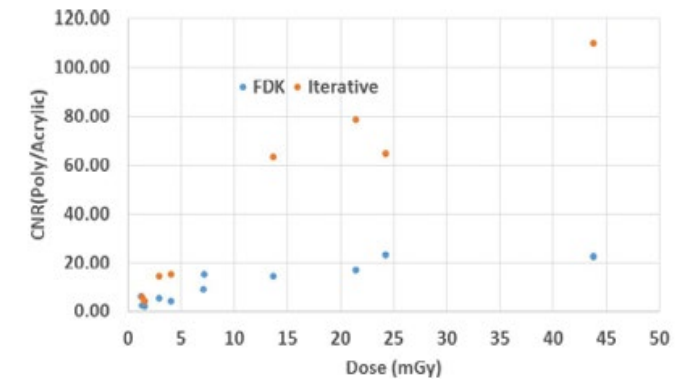
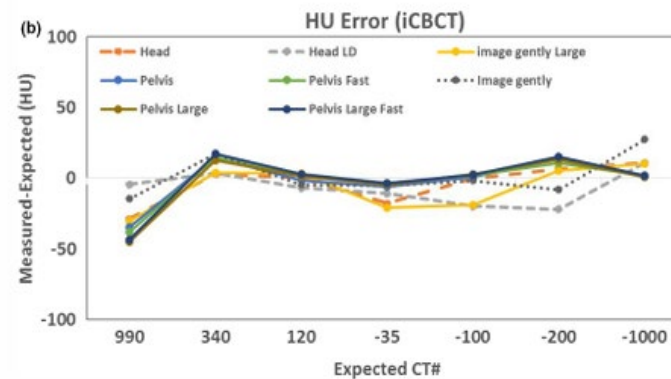
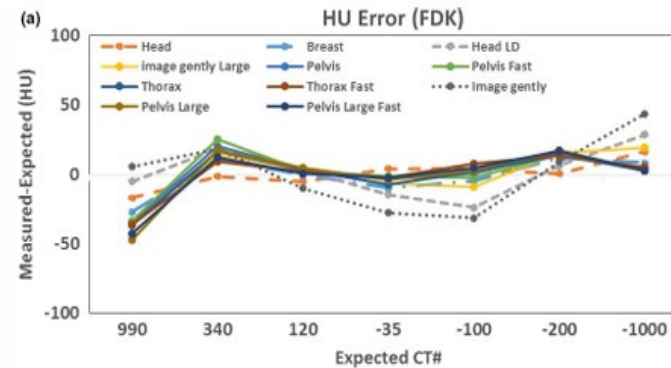
QA for subsystem – imaging system

What would you do to QA the
imaging system on this platform?



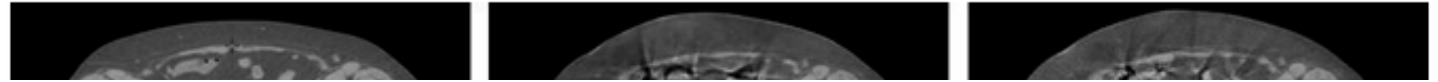
QA for subsystem – imaging system

- Phantom based QA: critical imaging metrics
 - Geometric accuracy
 - Image quality
 - HU accuracy
 - Contrast-to-noise ratio
 - Reconstruction accuracy
 - .etc.

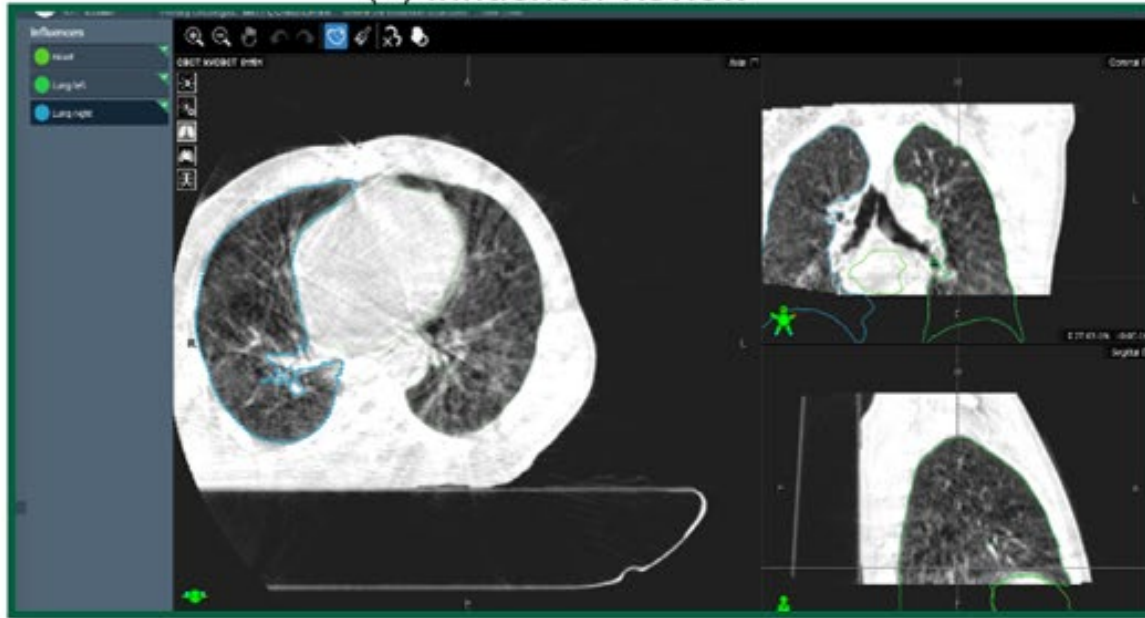


QA for subsystem – imaging system

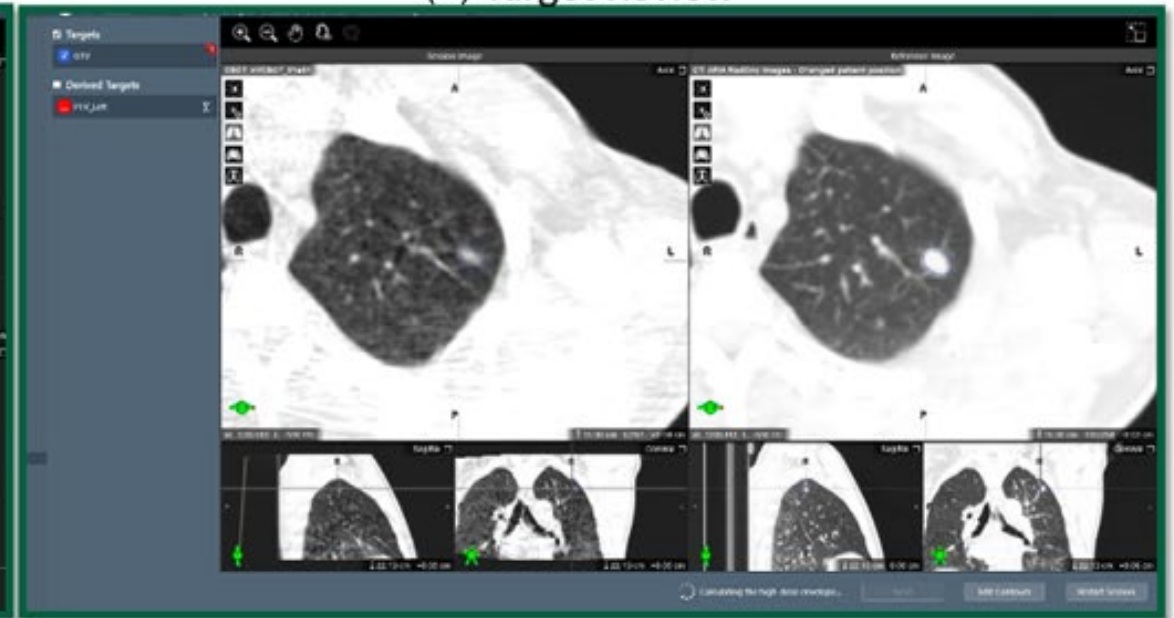
- Beyond Phantom QA: workflow and clinical usability -- adaptive-specific concerns
- Clinical contourability:



(a) Influencer Review



(b) Target Review



E2E testing of adaptive platform

- E2E testing is essential in ART to validate the entire adaptive chain
- Testing goal:
 - Stability (functional integrity, system communication, workflow readiness)
 - Quantify accuracy under realistic changes (geometric/dosimetric)
 - Establish ongoing QA baselines

Two Complementary E2E Testing Strategies

1. Dosimetry-Focused E2E

Confirms overall dose accuracy



- Phantoms with ion chambers, film, or detector arrays
- Deformable or moving targets to trigger meaningful adaptation
- Assess dose accuracy of adapted plans vs. reference

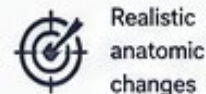
2. Workflow-Focused E2E

Stresses functionality & data integrity



- Anthropomorphic phantoms or vendor emulators (MR/CBCT replay)
- Test full chain: imaging → planning → adaptation → delivery
- Verify inter-system communication, data transfer, and workflow stability

Common Elements



Realistic anatomic changes



Geometric & dosimetric accuracy



Reproducible & periodic checks



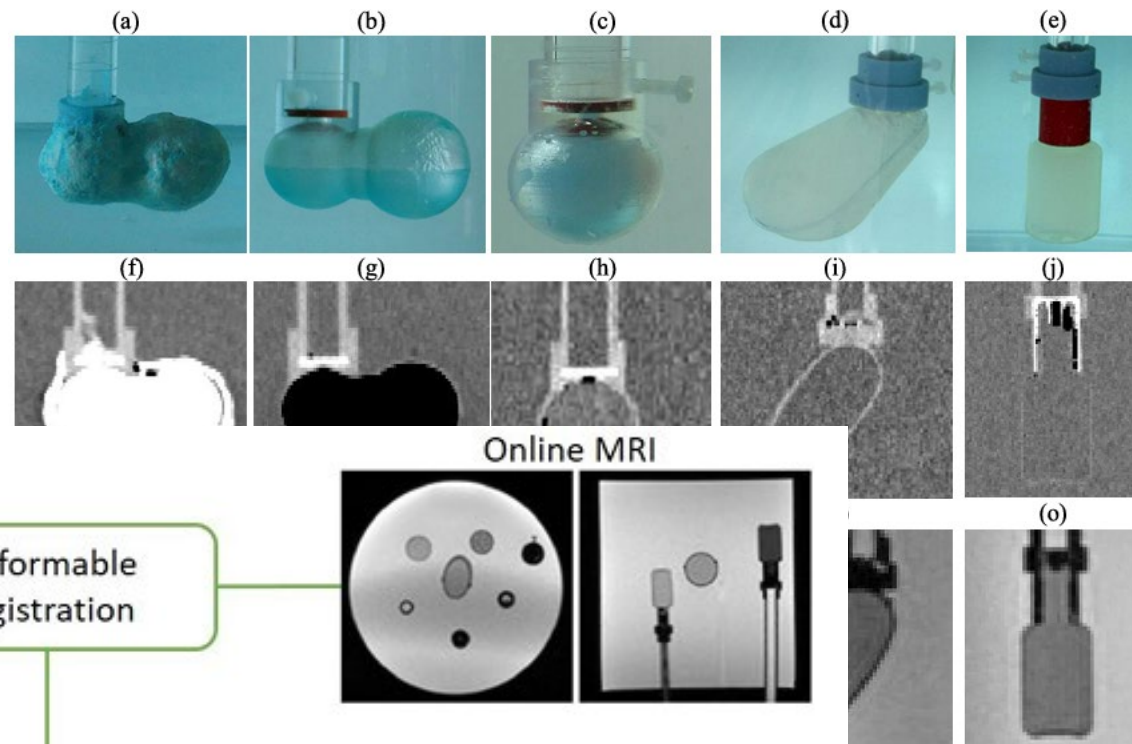
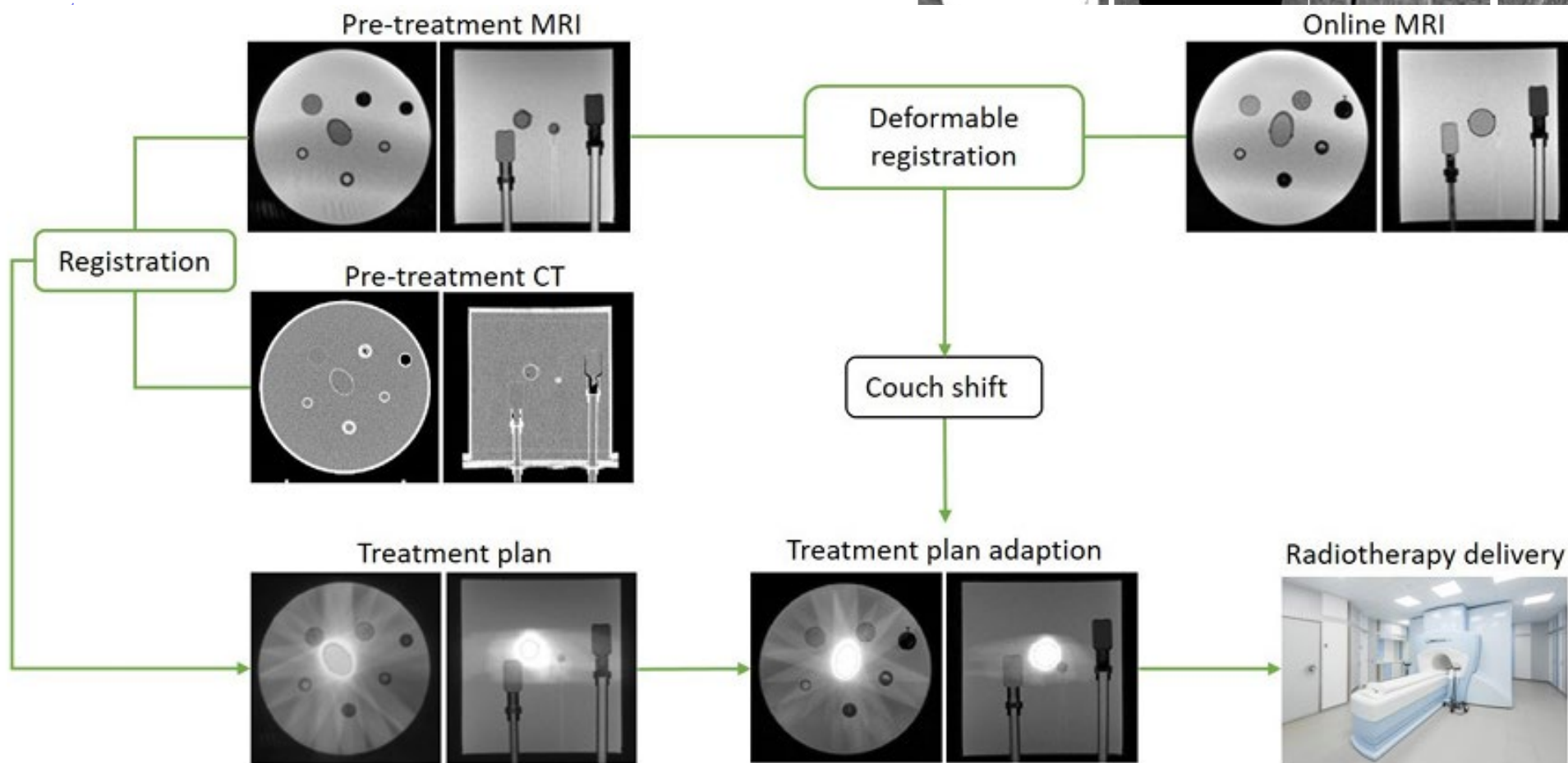
Action levels & risk-informed controls



Training, risk awareness & continuous improvement

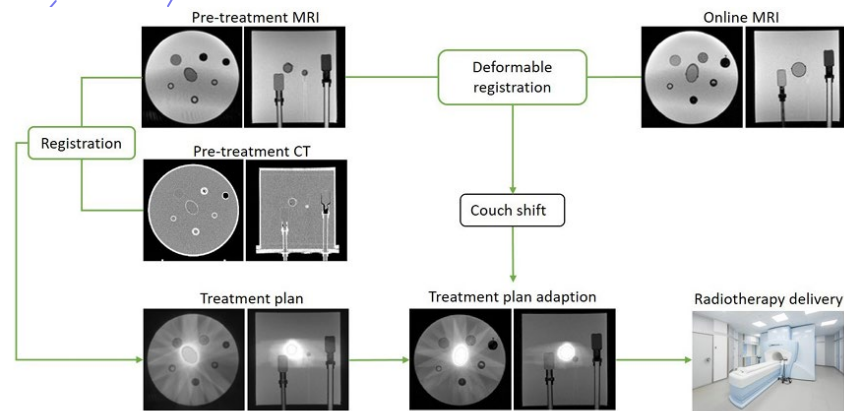
E2E testing examples

- E2E test for MRgART

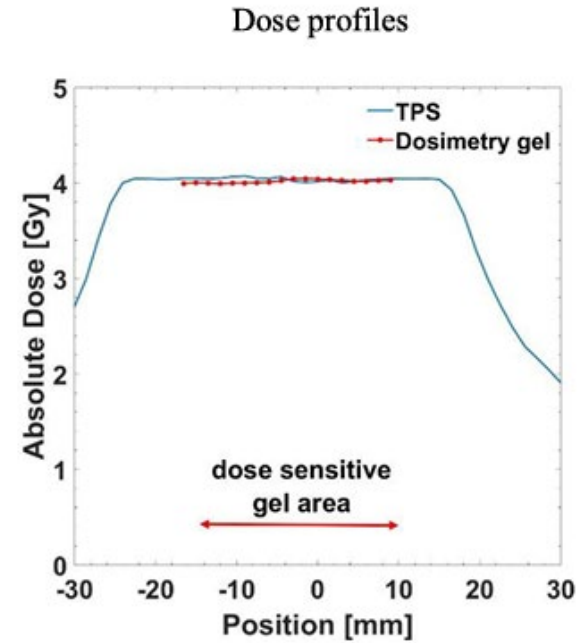


E2E testing examples

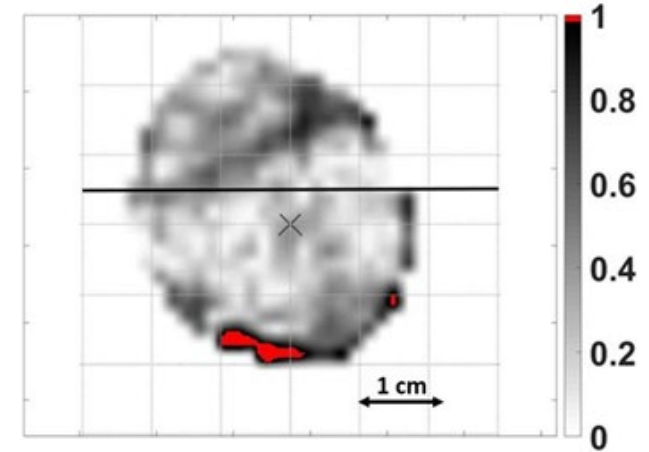
■ E2E test for MRgART



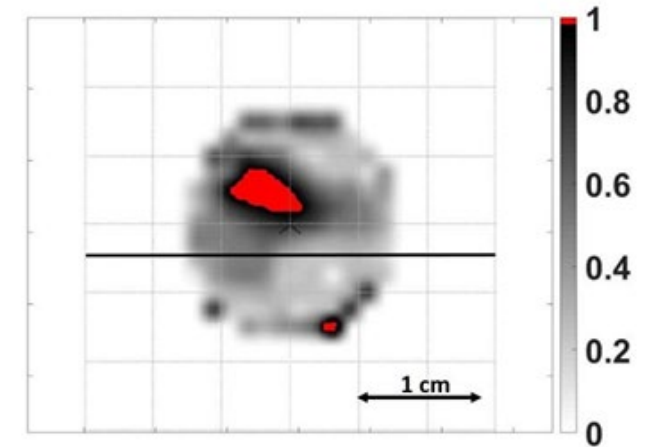
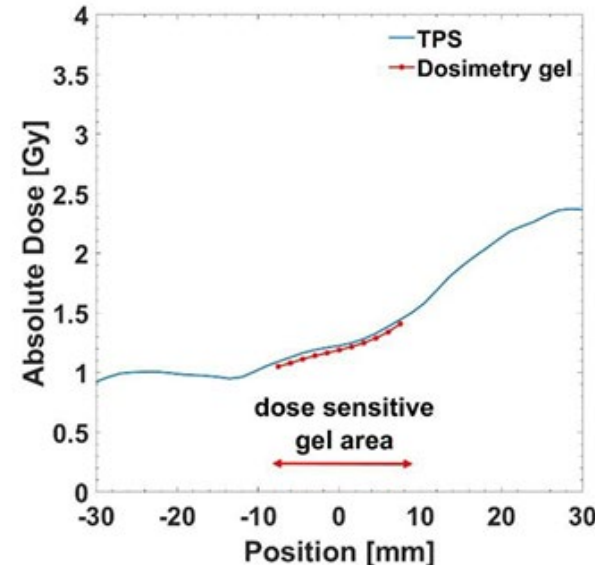
tumour



2D γ -map

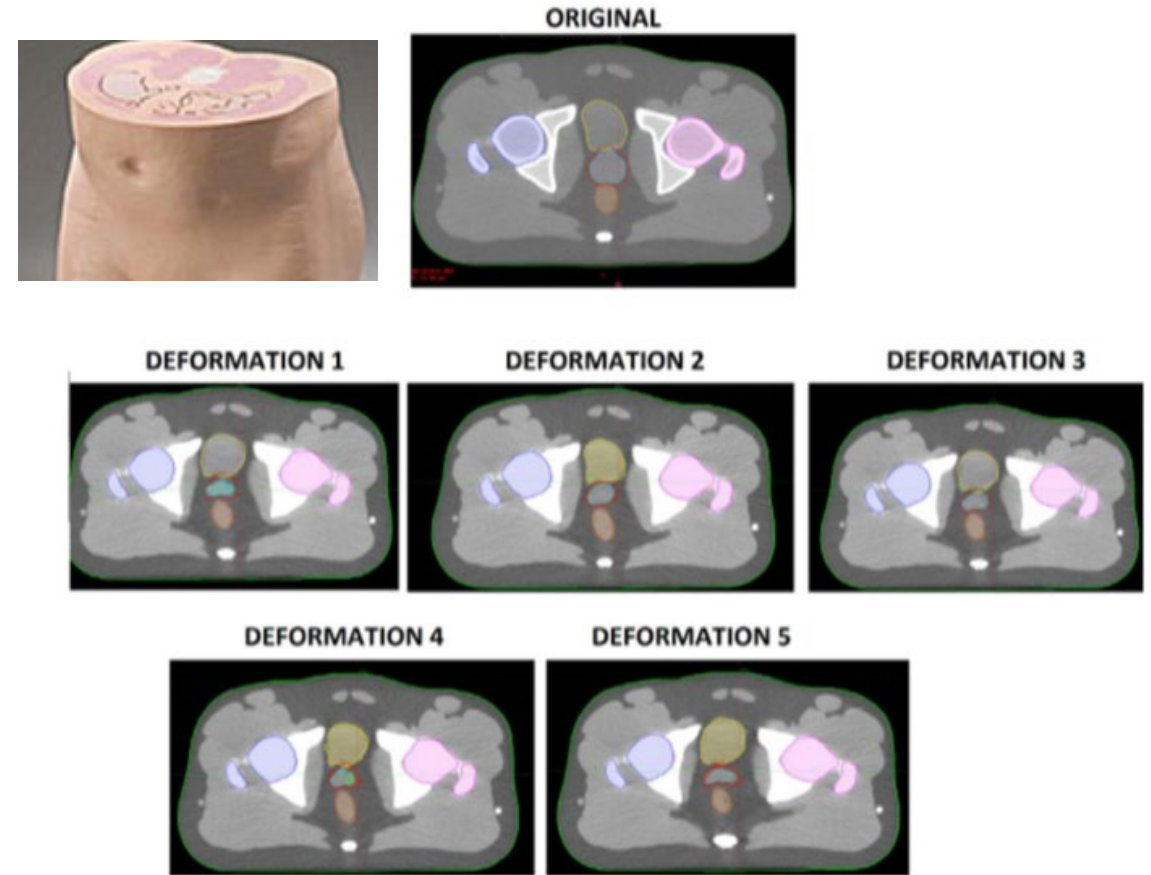


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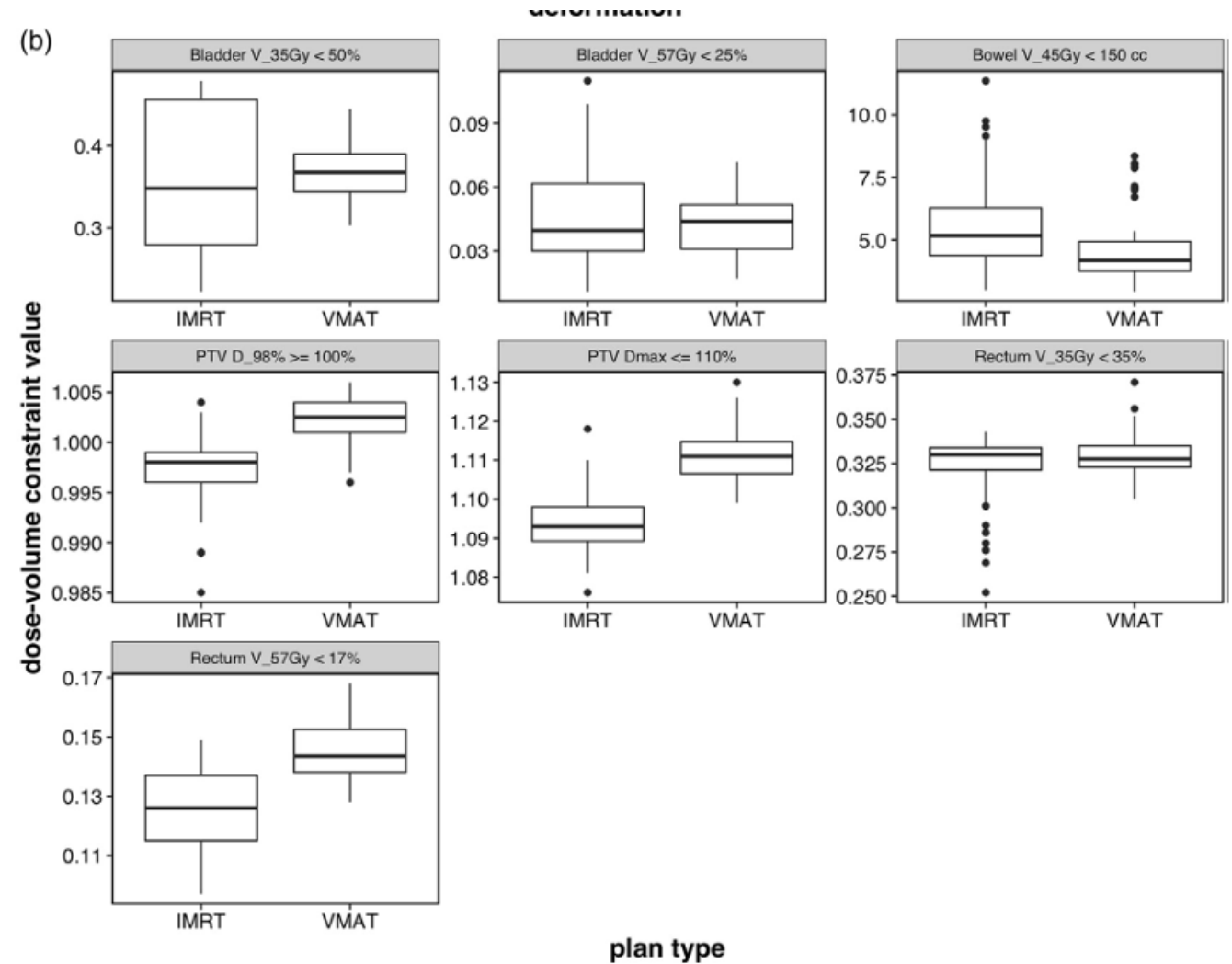
E2E testing examples

- E2E test for CBCTgART
 - Original CT is deformed (e.g. five configurations)
 - Each deformed CT use as reference CT for initial plan.
 - Phantom scanned on machine in clinical model.



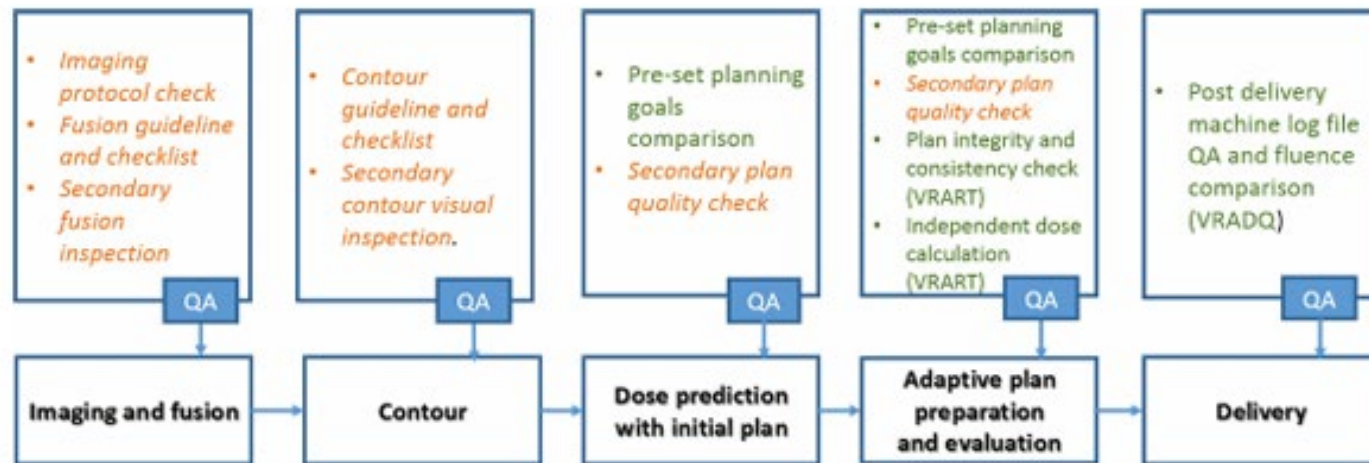
E2E testing examples

- E2E test for CBCTgART
 - Contour reproducibility
 - Manual contour reproducibility
 - Adapted plan reproducibility



QA for ART steps

- ART QA is workflow-integrated, Not an afterthought.
- QA/QC activities must be embedded at defined control points throughout the adaptive workflow.
- Essential components include: clear tolerance, acceptance criteria, clear team roles/responsibilities, time limits, benchmarking values, documentations, .etc





Discussion:

What QA would you do for adapted plan?



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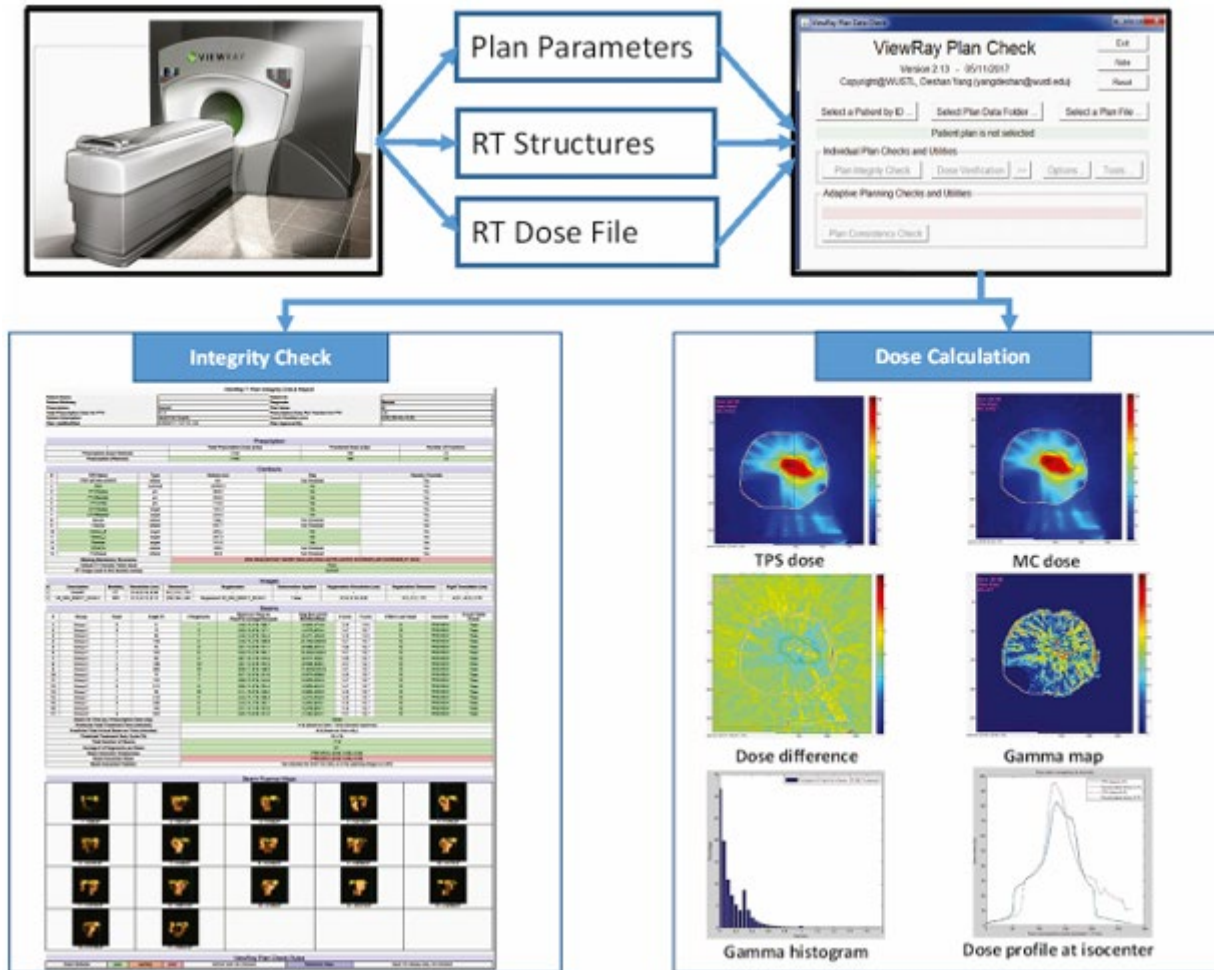


QA for adapted plan

- Two categories: plan quality review and integrity check.
- Plan quality review – focus on plan quality and clinical acceptance.
 - Robust initial plan with reusable objectives (*Mittauer et al*) and pre-set evaluation metrics (*PARTS Score Bohoudi et al*).
 - Focus on DVH, dose distribution, DV constraints.

QA for adapted plan

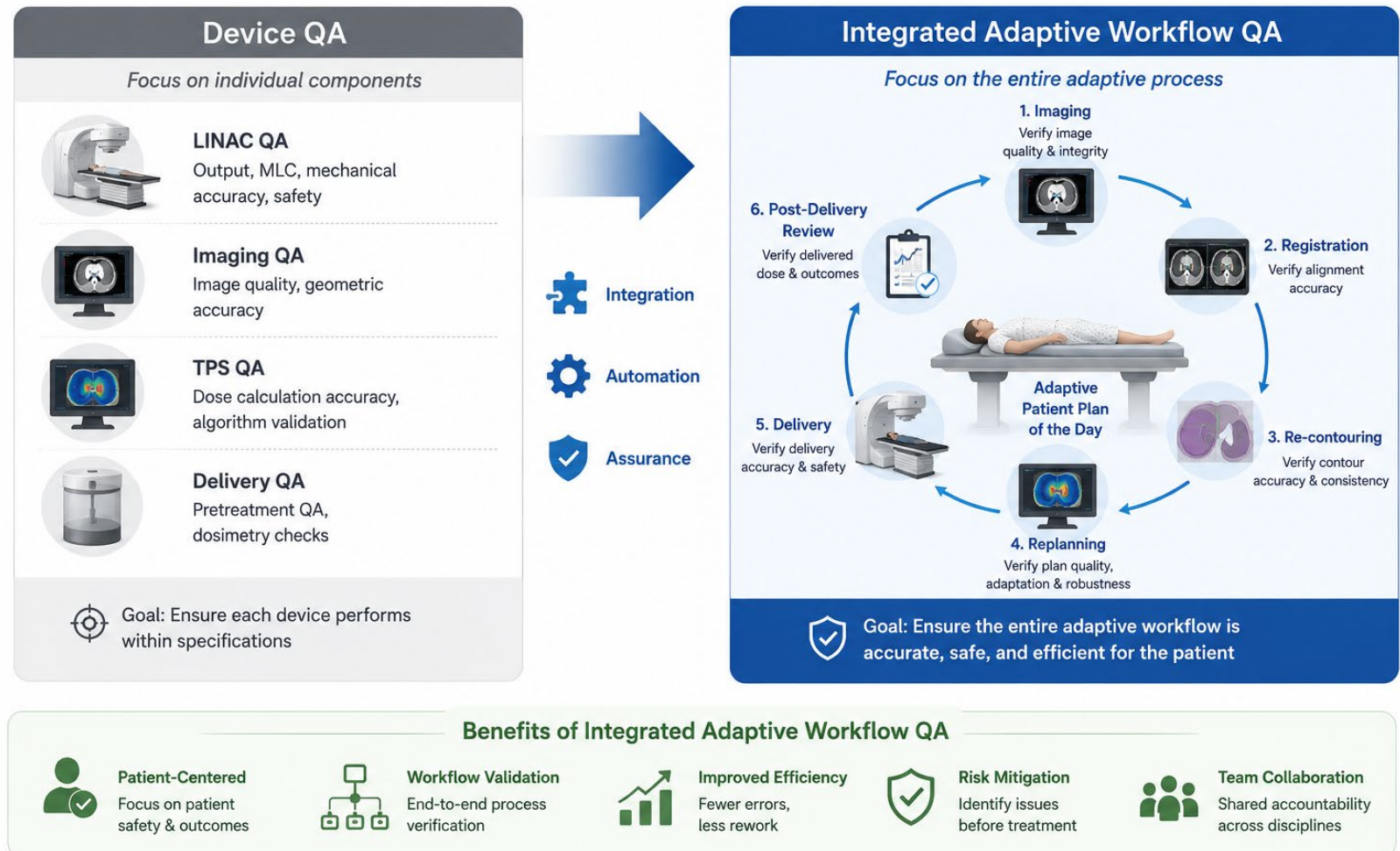
- Plan integrity review
 - Physics peer review with pre-set checklist (*Iqbal et al.*)
 - Plan or physics parameters: Electron-density assignment, MU reasonableness, delivery configuration, .etc.
 - Can be automated with rule-based checking.



Future QA direction

■ From Device QA -> Integrated Adaptive Workflow QA

- ❖ Fast, automated on table QA
- ❖ Beyond traditional Gamma analysis
- ❖ Expanded E2E testing
- ❖ Continuous QA monitoring



Future QA direction

- AI, Automation and standardization
 - ❖ Building a modern adaptive QA framework – FMEA driven (map, identify, design, deploy, continues improvement)
 - ❖ Automation will be central (rule based validation, hard-constraint verification, consistency check)
 - ❖ Emerging AI-Assisted QA (segmentation QA, dose accumulation QA, LLM-assisted plan/parameter triage)
 - ❖ Standardization across institutions

Acknowledgment

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- Vendor collaborators
 - Varian team and AIC
 - RefleXion team
 - Elekta team
 - United Imaging team
 - Sun Nuclear team
 - PTW team
- All collaborators over the years

Thank you for your attentions!

Questions?

cai.bin@mayo.edu



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